

# NEUROSCI 366

## Lecture 27

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### 系统巩固与泛化 / Systems Consolidation and Generalization

Source-aligned · Static · Searchable · Printable

#### Source files

- [Lecture27-SystemsConsolidation-Generalization.pdf](#)

**Production version:** 2026-08-22

**Reader profile:** neuroscience undergraduate education; mathematics scaffolded from high-school algebra/trigonometry to the formal computational-neuroscience level.

This companion preserves source-page order, distinguishes source material from necessary scaffolding and extensions, and places hints/answers after the lesson. It is a static PDF: no forms, JavaScript, hidden-answer widgets, passwords, or DRM.

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## 1 How to use this companion

1. 先用 5-minute diagnostic 暴露前置缺口；不要立即翻答案。
2. 按原笔记页序阅读 source-page units：先看原页，再读 clean reconstruction 与补全逻辑。
3. 遇到公式按“问题 → 符号/shape/units → assumptions → 一行一步推导 → intuition → biological meaning → sanity check → failure”阅读。
4. 完成 Cumulative Knowledge Check；只在需要时依次打开 Hint 1、2、3。
5. 至少隔一个 page break 后再核对 Answer Key，并记录错误类型：concept、symbol、algebra、assumption、correlation/causation。

**Source hierarchy.** 原笔记是内容权威；本书中的“【必要前置】/【补全推导】/【跨课连接】/【延伸】”是为了补齐数学或迁移，不替代源材料。无法确认的内容必须进入 Uncertainty Log。

## 2 Learning objectives and prerequisite dependency map

- 定义并区分：recent vs remote memory。
- 从第一原则推导：hippocampal replay。
- 解释几何/计算直觉：teacher-student-notebook。
- 连接生物学解释：pattern separation/completion。
- 诊断 assumptions 与 failure modes：memory error。
- 把概念迁移到新神经科学问题：generalization error。

### Dependency map

recent vs remote memory → hippocampal replay → teacher-student-notebook → pattern separation/completion → memory error → generalization error

### Core question

episodic memory 如何从 hippocampal one-shot notebook 向 neocortical predictive student 转移，为什么 standard consolidation 会过拟合 unpredictable details，而 Go-CLS 需按 predictability 停止？

## 3 Five-minute prerequisite diagnostic

1. systems consolidation 改变哪两个方面？（卡住时先看 Source-page unit 1，再看补全推导。）
2. standard account 中 hippocampal replay 的作用？（卡住时先看 Source-page unit 2，再看补全推导。）
3. Teacher、Student、Notebook 分别代表什么？（卡住时先看 Source-page unit 3，再看补全推导。）
4. pattern separation 的计算益处？（卡住时先看 Source-page unit 4，再看补全推导。）
5. pattern completion 的益处？（卡住时先看 Source-page unit 5，再看补全推导。）

**Scoring guide:** 4–5 confident answers → proceed; 2–3 → use the prerequisite labels; 0–1 → read the formula/notation sheet once before the source-aligned lesson.

## 4 Source-aligned lesson / 按原笔记页序

### 4.1 Lecture27-SystemsConsolidation-Generalization.pdf • p. 1

#### Memory consolidation:

The anatomical locus and psychological character of memory both change over time.

Our understanding of this has been heavily influenced by human neurology patients with cognitive impairments from brain damage

Watch video excerpt from patient E.P. (Youtube). E.P. has bilateral hippocampal damage, cannot form new explicit memories, and has amnesia for recent memories. Very old memories appear to be intact.

#### Standard theory of memory consolidation:

- Memories are initially stored in hippocampus.
- Over time, memories consolidate to distributed neocortical networks for long-term storage.

#### Hippocampal replay:

- "Place cells" report where animal is in environment.
- During rest, "place cells" replay trajectories taken through environment, compressed in time.
- This reactivates neocortical activity and could drive plasticity underlying systems consolidation.

#### Memory quality:

- We remember recent memories more vividly.
- Older memories are more gist like.
- Hippocampus: detailed episodic memories
- Neocortex: generalized semantic memories

**Original-note anchor.** [Source: Lecture27-SystemsConsolidation-Generalization.pdf, p. 1]

**Clean reconstruction.** Page 1: memory locus/quality over time, patient E.P., standard systems consolidation, hippocampal replay, episodic/semantic.

**What the note is saying.** 本页的中心对象是 recent vs remote memory 与 hippocampal replay。原笔记将它们放进以下链条: memory locus/quality over time、patient E.P.、standard systems consolidation、hippocampal replay、episodic/semantic。

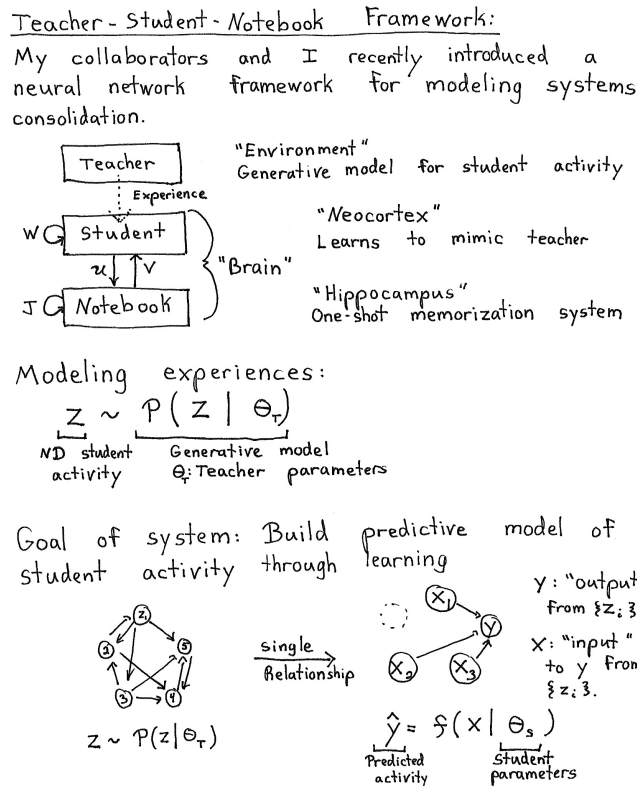
**Why it follows.** 解释 hippocampal replay、compressed trajectories 与 neocortical reactivation 之间的 hypothesized link。其逻辑后果是: 解释 memories 的 anatomical locus 与 psychological character 随时间改变; 准确呈现 patient E.P. 与 recent/remote amnesia evidence, 不夸大单案例。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

**Figure reading.** 先识别图中对象、箭头、参数和坐标系, 再问它表达的是定义、推导、模型预测还是经验观察; 示意图不提供未标注的定量精度。

#### Stop & Predict

systems consolidation 改变哪两个方面?

## 4.2 Lecture27-SystemsConsolidation-Generalization.pdf • p. 2



**Original-note anchor.** [Source: Lecture27-SystemsConsolidation-Generalization.pdf, p. 2]

**Clean reconstruction.** Page 2: Teacher-Student-Notebook framework, generative teacher, neocortical student, hippocampal one-shot notebook.

**What the note is saying.** 本页的中心对象是 teacher-student-notebook 与 pattern separation/completion。原笔记将它们放进以下链条: Teacher-Student-Notebook framework、generative teacher、neocortical student、hippocampal one-shot notebook。

**Why it follows.** 逐项重构 Teacher-Student-Notebook: teacher/environment generative model, student/neocortex predictive learner、notebook/hippocampus one-shot autoassociative memory。其逻辑后果是: 解释 hippocampal replay、compressed trajectories 与 neocortical reactivation 之间的 hypothesized link。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

**Figure reading.** 先识别图中对象、箭头、参数和坐标系, 再问它表达的是定义、推导、模型预测还是经验观察; 示意图不提供未标注的定量精度。

### Stop & Predict

standard account 中 hippocampal replay 的作用?

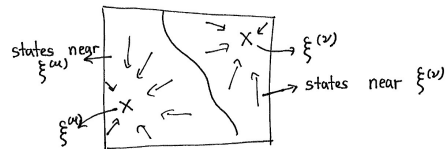
## 4.3 Lecture27-SystemsConsolidation-Generalization.pdf • p. 3

The notebook aids student learning by providing it with memory.

Want to store:  $\{Z^{(\mu)}, \xi^{(\mu)}\}, \mu=1, \dots, P$   
student activity during encoding      notebook activity during encoding

Achieved by modeling notebook as an auto-associative Hopfield model (last lecture). [Buhmann et al variety]

- Pattern separation:  $\xi^{(\mu)}$  is random and independent of  $Z^{(\mu)}$ . This reduces interference between memories and facilitates Hebbian learning of  $J$ .
- Pattern completion: Attractor dynamics drive notebook into a memorized pattern,  $\xi^{(\mu)}$ , through  $J$ .



- Hebbian learning between student & notebook permits student reactivation by notebook

$$\tilde{Z}_{i\mu} = \sum_{j=1}^M V_{ij} \xi_{j\mu}$$

$$V_{ij} = \frac{1}{Ma(1-a)} \sum_{\mu=1}^P Z_{i\mu} (\xi_{j\mu} - a)$$

$$\Delta V_{ij}^{(\mu)} = \frac{1}{Ma(1-a)} Z_{i\mu} (\xi_{j\mu} - a)$$

and notebook seeding by student.

**Original-note anchor.** [Source: Lecture27-SystemsConsolidation-Generalization.pdf, p. 3]

**Clean reconstruction.** Page 3: autoassociative notebook, pattern separation/completion, student-notebook Hebbian interactions.

**What the note is saying.** 本页的中心对象是 memory error 与 generalization error。原笔记将它们放进以下链条：autoassociative notebook、pattern separation/completion、student-notebook Hebbian interactions。

**Why it follows.** 解释 notebook 的 pattern separation、pattern completion、Hebbian student↔notebook interaction 与 seeding。其逻辑后果是：逐项重构 Teacher-Student-Notebook: teacher/environment generative model、student/neocortex predictive learner、notebook/hippocampus one-shot autoassociative memory。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

**Figure reading.** 先识别图中对象、箭头、参数和坐标系，再问它表达的是定义、推导、模型预测还是经验观察；示意图不提供未标注的定量精度。

### Stop & Predict

Teacher、Student、Notebook 分别代表什么？

## 4.4 Lecture27-SystemsConsolidation-Generalization.pdf • p. 4

Linear model: I'll present results from an analytically tractable linear model of this framework.

$$\text{Teacher: } y = \underbrace{\sum_{i=1}^N \bar{w}_i x_i}_{\text{Teacher weights}} + \underbrace{\epsilon}_{\text{Noise}}$$

$$\text{Student: } \hat{y} = \sum_{i=1}^N \underbrace{w_i}_{\text{Student weights}} x_i$$

Memory assessed with  $P$  stored patterns.

$$e_m^2 = \frac{1}{P} \sum_{\mu=1}^P (y_\mu - \hat{y}_\mu)^2 = \frac{1}{P} \sum_{\mu=1}^P (y_\mu - \sum_{i=1}^N w_i \tilde{x}_{i\mu})^2$$

Recalled  $x_{i\mu}$   
 $\approx x_{i\mu}$

Generalization assessed over all possibilities

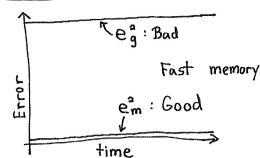
$$e_g^2 = \underbrace{\langle (y - \hat{y})^2 \rangle}_{\text{Average over } P(z|\theta_T)} = \langle (y - \sum_{i=1}^N w_i x_i)^2 \rangle$$

Standard model of systems consolidation. Adjust  $w$  to minimize  $e_m^2$  and transfer memories.

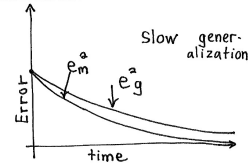
$$\eta \frac{dw_i}{dt} = -\frac{\partial}{\partial w_i} e_m^2 = \frac{2}{P} \sum_{\mu=1}^P e_{\mu} x_{i\mu}$$

First assume  $\epsilon=0$ . Reveals complementary learning systems (idea from McClelland, McNaughton, O'Reilly)

Notebook



Student



**Original-note anchor.** [Source: Lecture27-SystemsConsolidation-Generalization.pdf, p. 4]

**Clean reconstruction.** Page 4: linear teacher/student model, memory error, generalization error, standard complementary learning systems.

**What the note is saying.** 本页的中心对象是 overfit/stop 与 Go-CLS lesion predictions。原笔记将它们放进以下链条: linear teacher/student model、memory error、generalization error、standard complementary learning systems。

**Why it follows.** 从 linear teacher/student model 定义 memory error 与 population generalization error; 说明 fast notebook memory 与 slow student generalization 的 tradeoff。其逻辑后果是: 逐项重构 Teacher-Student-Notebook: teacher/environment generative model、student/neocortex predictive learner、notebook/hippocampus one-shot autoassociative memory。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

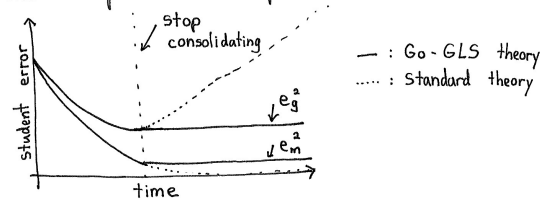
**Figure reading.** 先识别图中对象、箭头、参数和坐标系, 再问它表达的是定义、推导、模型预测还是经验观察; 示意图不提供未标注的定量精度。

### Stop & Predict

pattern separation 的计算益处?

## 4.5 Lecture27-SystemsConsolidation-Generalization.pdf • p. 5

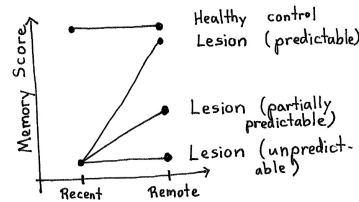
If  $\epsilon \neq 0$ , maintaining this generalization benefit requires student learning to stop before it overfits unpredictable components of experience.



Generalization-optimized complementary learning systems (Go-CLS) predicts that unpredictable memory components do not consolidate. A signal-to-noise ratio controls the extent of memory transfer

$$\lambda = \frac{\langle (\bar{w} \cdot x)^2 \rangle}{\langle \epsilon^2 \rangle}$$

$\lambda$ : degree of predictability



Hippocampus is required to recall fully detailed episodic memories containing unpredictable elements.

Neocortex integrates memories to extract the predictable semantic content.

**Original-note anchor.** [Source: Lecture27-SystemsConsolidation-Generalization.pdf, p. 5]

**Clean reconstruction.** Page 5: overfitting unpredictable components, stopping consolidation, Go-CLS, predictability/SNR, lesion predictions.

**What the note is saying.** 本页的中心对象是 recent vs remote memory 与 hippocampal replay。原笔记将它们放进以下链条: overfitting unpredictable components, stopping consolidation, Go-CLS, predictability/SNR, lesion predictions。

**Why it follows.** 讲解 generalization-optimized CLS (Go-CLS)、predictability/SNR-controlled transfer、lesion predictions, 以及为什么 unpredictable episodic details remain hippocampus-dependent。其逻辑后果是: 解释 standard CLS 在 noise/unpredictable components 下可能 overfit; 推导/解释停止巩固的必要性。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

**Figure reading.** 先识别图中对象、箭头、参数和坐标系, 再问它表达的是定义、推导、模型预测还是经验观察; 示意图不提供未标注的定量精度。

### Stop & Predict

pattern completion 的益处?



## 5 补全推导与数学支架 / Derivations

### 5.1 empirical observations 与 standard consolidation

**【必要前置】** recent/remote amnesia 与 memory quality 如何组织理论？

recent: hippocampus-dependent, detailed → remote: distributed cortical, gist-like

1. patient evidence (如 E.P.) 显示 bilateral hippocampal damage 严重影响 new explicit memories, 并常有 temporal gradient。
2. place-cell replay 在 rest/sleep 中以 compressed sequences 重现 trajectories。
3. standard account 认为 replay/reactivation 驱动 neocortical plasticity。
4. single cases 与 correlations 不能单独证明唯一 mechanism。

**Geometric/computational intuition.** 记忆不是简单从一个硬盘复制到另一个；representation 与 detail 也会改变。

**Biological interpretation / failure condition.** multiple trace/transformation views 与 task-dependent hippocampal roles 提供替代理论。

#### Micro-check

pattern separation 的计算益处？

## 5.2 Teacher–Student–Notebook formal roles

**【补全推导】** 三个系统分别解决什么 computational problem?

$$z \sim P(z \mid \theta_T)$$

$$\hat{y} = f(x; \theta_S)$$

$$\{z^\mu\}_{\mu=1}^P \leftrightarrow \{\xi^\mu\}_{\mu=1}^P$$

1. Teacher/environment 生成 structured experiences。
2. Student/neocortex 用 slow learning 提取可预测 input-output relation。
3. Notebook/hippocampus 用 autoassociative one-shot storage 保存 episodes。
4. notebook pattern separation 减少 memories interference, pattern completion 支持 replay/seed-ing student。

**Geometric/computational intuition.** fast memory 解决单次保存, slow learner 解决跨 experiences 泛化。

**Biological interpretation / failure condition.** 三个角色是 formal abstractions, 不等于每个 anatomical region 只执行单一功能。

### Micro-check

pattern completion 的益处?

### 5.3 linear teacher/student 的 memory 与 generalization error

**【跨页连接】** 为什么训练误差下降不保证 population prediction 变好？

$$y = \mathbf{w}_T^T \mathbf{x} + \epsilon$$

$$e_m = \frac{1}{P} \sum_{\mu=1}^P (y_\mu - \mathbf{w}_S^T \mathbf{x}_\mu)^2$$

$$e_g = E_{x,\epsilon} [(y - \mathbf{w}_S^T \mathbf{x})^2]$$

1. memory/training error 只在 notebook stored examples 上评估。
2. generalization error 对 teacher distribution 的新 samples 取 expectation。
3. fast notebook 可令 stored pattern recall 很好，即  $e_m$  低。
4. student 需要多 examples 才能估 teacher signal，过度拟合  $\epsilon$  会使  $e_g$  上升。

**Geometric/computational intuition.** 记住每个 episode 与学到可预测规律是不同 objectives。  
**Biological interpretation / failure condition.** nonlinear/high-dimensional teacher 下 curves 取决于 model capacity、regularization 与 sample size。

#### Micro-check

memory error 与 generalization error 区别？

## 5.4 Go-CLS stopping 与 predictability

**【模型边界】** 何时应停止 hippocampus→cortex consolidation?

$$\lambda_s = \frac{E[(\mathbf{w}_T^T \mathbf{x})^2]}{E[\epsilon^2]} \quad (\text{predictability/SNR})$$

$$t^* = \arg \min_t e_g(t)$$

1. 早期 transfer 学 signal, e\_g 下降。
2. 继续 replay 后, student 开始拟合 notebook 中的 unpredictable  $\epsilon$ , e\_g 回升。
3. Go-CLS 在 minimum generalization error 附近停止/衰减 transfer。
4. 高  $\lambda_s$  memories 更可 corticalize; 低  $\lambda_s$  detail 保持 hippocampal dependence。

**Geometric/computational intuition.** 最优 consolidation 不是复制到零 training error, 而是最大化未来 prediction。

**Biological interpretation / failure condition.** brain 如何估计 predictability/停止时机是 model prediction, 需实验检验。

### Micro-check

为何 standard unlimited transfer 可过拟合?

## 6 Cross-page synthesis / 跨页因果与数学链

systems consolidation 同时改变 memory 的 anatomical dependence 与 psychological character: recent memories 更 vivid/episodic, remote memories 更 gist-like/semantic. Teacher-Student-Notebook framework 把 environment/teacher 看作 generative source, neocortical student 慢学 predictive structure, hippocampal notebook 快速存 one-shot episodes 并 replay. 标准持续 transfer 会把 unpredictable noise 也写入 student, 降低 generalization; Go-CLS 根据 signal predictability/SNR 控制 transfer 并在最优点停止, 因此 unpredictable episodic detail 保持 hippocampus-dependent.

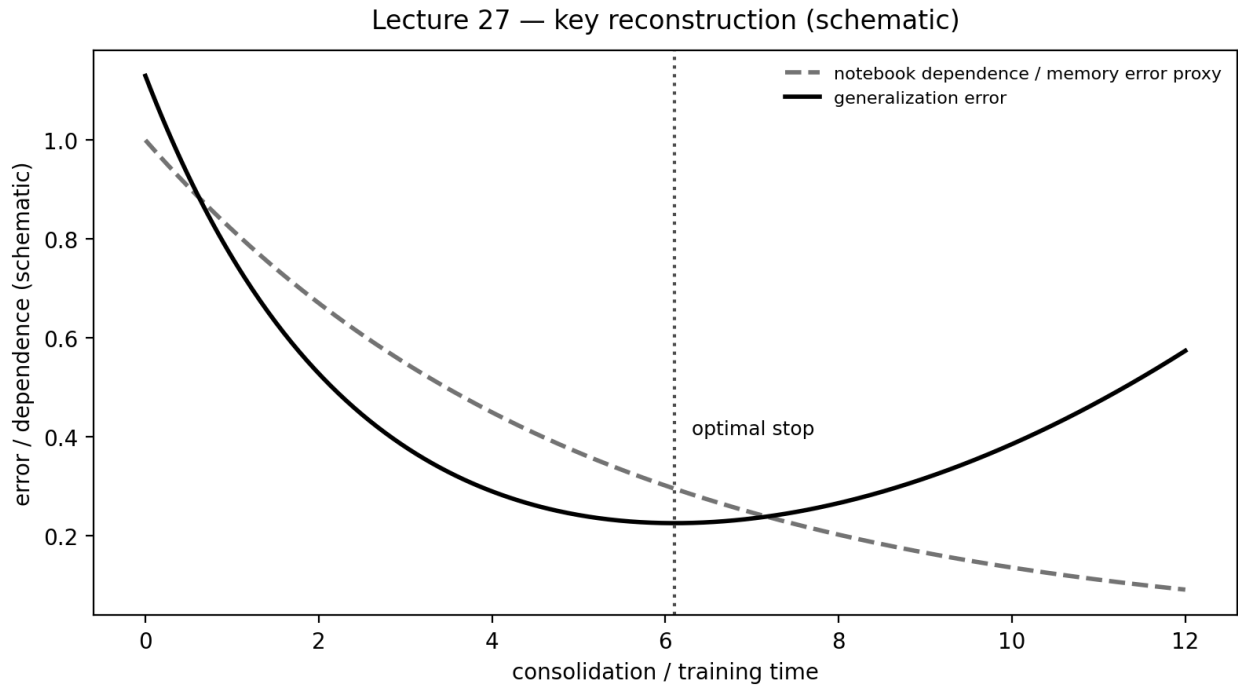


Figure note: schematic reconstruction; it encodes qualitative relations from the lesson and does not invent precise empirical data points.

## 7 Worked examples and transfer

### 7.1 signal vs episode noise

**【原课/核心例题】** Prompt. 多次餐馆经历中, 菜单结构稳定, 但某天邻桌说的一句话不可预测。

**Worked solution.** neocortex student 可学菜单/语义规律; 具体一句话作为 episode detail 低 predictability, 长期可能仍依 hippocampal trace。

**Check.** gist/detail 差异不是简单时间长短, 而与跨 episodes 可预测 structure 有关。

### 7.2 lesion prediction

**【跨课连接】** Prompt. memory component 高 predictability vs low predictability, 随时间做 hippocampal lesion。

**Worked solution.** 高 predictability component 随 consolidation 变得较 lesion-resistant; 低 predictability component 远期仍受 lesion 强烈影响; partial predictability 居中。

**Check.** 这是 Go-CLS prediction, 不应写成已无争议事实。

### Micro-check

**Explain in your own words.** 用不超过 120 字解释: 本课从 source page 1 到最后一页, 究竟把哪一个输入、状态、统计量或学习规则变成了可检验 prediction?

**Self-reflection.** 你最可能犯的是 concept、symbol、algebra、assumption 还是 causality error? 写出一个具体例子。

## 8 Formula and notation sheet

### 1. linear teacher

$$y = w_T^T x + \epsilon$$

Conditions / conventions: signal+unpredictable noise

### 2. student predictor

$$\hat{y} = w_S^T x$$

Conditions / conventions: linear model

### 3. memory/training error

$$e_m = P^{-1} \sum (y_\mu - \hat{y}_\mu)^2$$

Conditions / conventions: stored patterns

### 4. population generalization error

$$e_g = E[(y - \hat{y})^2]$$

Conditions / conventions: new experiences

### 5. predictability/SNR

$$\lambda_s = \text{Var}(\text{signal}) / \text{Var}(\text{noise})$$

Conditions / conventions: finite variances

### 6. optimal stopping

$$t^* = \arg \min e_g(t)$$

Conditions / conventions: learning curve measurable

### 7. consolidation channel

replay → student updates

Conditions / conventions: formal model hypothesis

## 9 Bilingual glossary

| 中文        | English / symbol                    | Operational meaning                                                          |
|-----------|-------------------------------------|------------------------------------------------------------------------------|
| 系统巩固      | systems consolidation               | memory dependence/representation 在 brain systems 间随时间变化。                     |
| 情景记忆      | episodic memory                     | 特定事件、时间、地点与细节的 memory。                                                       |
| 语义记忆      | semantic memory                     | 较抽象、generalized knowledge。                                                   |
| 重放        | replay                              | 休息/睡眠中对 experience-related neural sequences 的再激活。                            |
| 互补学习系统    | complementary learning systems, CLS | fast hippocampal 与 slow cortical learning 的 framework。                       |
| 教师-学生-笔记本 | Teacher-Student-Notebook            | generative source、slow predictor、one-shot associative memory 的 formal roles。 |
| 模式分离      | pattern separation                  | 把相似 experiences 映射为较不同 memory codes。                                         |
| 模式补全      | pattern completion                  | 由 partial cue 恢复 stored episode。                                             |
| 记忆误差      | memory error                        | 在 stored examples 上的 reconstruction/prediction error。                        |

| 中文     | English / symbol             | Operational meaning                                       |
|--------|------------------------------|-----------------------------------------------------------|
| 泛化误差   | generalization error         | 对新 samples 的 expected prediction error。                   |
| 过拟合    | overfitting                  | 拟合 sample-specific noise 导致新数据表现变差。                       |
| Go-CLS | generalization-optimized CLS | 以 generalization/predictability 控制 consolidation 的 model。 |

## 10 Common traps, assumptions, and limitations

1. 把 patient E.P. 单一案例当作全部 consolidation 机制证明。
2. recent=hippocampus、remote=neocortex 写成绝对二分。
3. replay correlation 直接等于 causal synaptic transfer。
4. 把 Teacher/Student/Notebook 当作一一解剖标签而非 formal roles。
5. memory error 与 generalization error 混为同一指标。
6. training error 越低就认为 consolidation 越好。
7. 把 Go-CLS predictions 写成已证实事实。

**Course-specific risk boundary.** 明确区分 empirical observations、standard theory、Teacher-Student-Notebook formal model 和 Go-CLS predictions；不要把模型预测写成已证实事实。



## 11 Cumulative Knowledge Check

Do not turn to the Hint Bank until you have written a first attempt.

1. systems consolidation 改变哪两个方面？

---

2. standard account 中 hippocampal replay 的作用？

---

3. Teacher、Student、Notebook 分别代表什么？

---

4. pattern separation 的计算益处？

---

5. pattern completion 的益处？

---

6. memory error 与 generalization error 区别？

---

7. 为何 standard unlimited transfer 可过拟合？

---

8. Go-CLS 的 stopping criterion？

---

9. 高 predictability memory component 的 lesion prediction？

---

10. 低 predictability detail 的 prediction？

---

11.  $\lambda_s$  大表示什么？

---

12. 为什么模型 prediction 不能写成事实？

---

13. 公式表中的 “linear teacher” 解决什么问题？写出适用条件与一个 sanity check。

---

14. 公式表中的 “student predictor” 解决什么问题？写出适用条件与一个 sanity check。

---

15. 公式表中的 “memory/training error” 解决什么问题？写出适用条件与一个 sanity check。

- 
16. 公式表中的 “population generalization error” 解决什么问题？写出适用条件与一个 sanity check。
- 
17. 公式表中的 “predictability/SNR” 解决什么问题？写出适用条件与一个 sanity check。
- 
18. 公式表中的 “optimal stopping” 解决什么问题？写出适用条件与一个 sanity check。
-

## 12 Hint Bank

1. **Hint 1.** 先识别对象、变量与假设。  
**Hint 2.** 写出最相关的定义或方程，再代入。  
**Hint 3.** 检查单位、shape、符号与极限情况。
2. **Hint 1.** 先识别对象、变量与假设。  
**Hint 2.** 写出最相关的定义或方程，再代入。  
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12. **Hint 1.** 先识别对象、变量与假设。  
**Hint 2.** 写出最相关的定义或方程，再代入。  
**Hint 3.** 检查单位、shape、符号与极限情况。
13. **Hint 1.** 先从“linear teacher”对应的变量关系入手。  
**Hint 2.** 把条件“signal+unpredictable noise”逐字转换为 assumption。  
**Hint 3.** 写公式，再检查至少一种极端参数或单位。
14. **Hint 1.** 先从“student predictor”对应的变量关系入手。  
**Hint 2.** 把条件“linear model”逐字转换为 assumption。  
**Hint 3.** 写公式，再检查至少一种极端参数或单位。

15. **Hint 1.** 先从 “memory/training error” 对应的变量关系入手。  
**Hint 2.** 把条件 “stored patterns” 逐字转换为 assumption。  
**Hint 3.** 写公式, 再检查至少一种极端参数或单位。
16. **Hint 1.** 先从 “population generalization error” 对应的变量关系入手。  
**Hint 2.** 把条件 “new experiences” 逐字转换为 assumption。  
**Hint 3.** 写公式, 再检查至少一种极端参数或单位。
17. **Hint 1.** 先从 “predictability/SNR” 对应的变量关系入手。  
**Hint 2.** 把条件 “finite variances” 逐字转换为 assumption。  
**Hint 3.** 写公式, 再检查至少一种极端参数或单位。
18. **Hint 1.** 先从 “optimal stopping” 对应的变量关系入手。  
**Hint 2.** 把条件 “learning curve measurable” 逐字转换为 assumption。  
**Hint 3.** 写公式, 再检查至少一种极端参数或单位。

### 13 Complete Answer Key with reasoning

1. memory 的 anatomical dependence/locus 与 psychological quality/detail。
2. 重新激活 neocortical patterns、驱动 slow cortical learning 的候选机制。
3. environment generative model、neocortical predictor、hippocampal one-shot associative memory。
4. 减少相似 episodes 的 interference。
5. partial cue 恢复完整 episode/replay seed。
6. 前者在 stored samples; 后者在新 samples/distribution。
7. student 开始学习 notebook 中不可预测 episode noise。
8. 在 generalization error minimum/由 predictability 控制的时点停止或减弱 transfer。
9. 随时间更 cortical/less hippocampus-dependent。
10. 即使 remote 仍较 hippocampus-dependent。
11. signal variance 相对 unpredictable noise 大, 可泛化 structure 强。
12. 需独立 behavioral/neural/lesion experiments 验证, 且存在替代理论。
13. 适用条件/约定: signal+unpredictable noise。sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$y = w_T^T x + \epsilon$$

14. 适用条件/约定: linear model。sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$\hat{y} = w_S^T x$$

15. 适用条件/约定: stored patterns。sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$e_m = P^{-1} \sum (y_\mu - \hat{y}_\mu)^2$$

16. 适用条件/约定: new experiences。sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$e_g = E[(y - \hat{y})^2]$$

17. 适用条件/约定: finite variances。sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$\lambda_s = \text{Var}(\text{signal}) / \text{Var}(\text{noise})$$

18. 适用条件/约定: learning curve measurable。sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$t^* = \arg \min e_g(t)$$

## 14 Source-page concordance and coverage audit

| Source file                                       | Page | Coverage status                                                                     |
|---------------------------------------------------|------|-------------------------------------------------------------------------------------|
| Lecture27-SystemsConsolidation-Generalization.pdf | 1    | Main source-page unit: thumbnail, reconstruction, explanation, prediction question. |
| Lecture27-SystemsConsolidation-Generalization.pdf | 2    | Main source-page unit: thumbnail, reconstruction, explanation, prediction question. |
| Lecture27-SystemsConsolidation-Generalization.pdf | 3    | Main source-page unit: thumbnail, reconstruction, explanation, prediction question. |
| Lecture27-SystemsConsolidation-Generalization.pdf | 4    | Main source-page unit: thumbnail, reconstruction, explanation, prediction question. |
| Lecture27-SystemsConsolidation-Generalization.pdf | 5    | Main source-page unit: thumbnail, reconstruction, explanation, prediction question. |

## 15 Errata, uncertainty log, and external endnotes

### 15.1 Errata / source cautions

1. Page 1 的 E.P. evidence 应作为 influential case evidence, 不夸大单案例普遍性。
2. Pages 2–5 的 Teacher–Student–Notebook/Go-CLS 是 formal model 与 predictions, 需与 empirical observations 分层。
3. Page 5 的 predictability ratio 是 signal-to-noise style quantity; 具体 normalization 依模型 convention。

### 15.2 Uncertainty log

No unresolved handwriting uncertainty changes a core definition, equation, figure interpretation, or answer in this companion. Where handwriting was potentially ambiguous, the source-page image is preserved and the reconstruction avoids claiming unverified detail.

### 15.3 External endnotes

No external web or textbook source was used to replace the uploaded notes. Added material is limited to necessary mathematical scaffolding, explicit assumptions, standard sanity checks, and clearly labeled transfer examples.

### 15.4 Final self-audit

All source pages listed above were visually inspected. Equations were checked for symbol definitions, units/shapes, assumptions, algebraic transitions, limiting cases, and failure conditions. The PDF is static and contains no interactive form fields or scripts.